

INFORMATION TECHNOLOGY DEPARTMENT

FINAL PROJECT

SY. 2024-2025 – 1ST SEMESTER
NET 313 – Advance Networking

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	Year & Section	BSIT 3-L	Date		

Project Title: Advanced Network Architecture with DHCP and EIGRP Optimization

Objective:

Students will design, implement, and optimize a network that integrates DHCP and EIGRP, analyze its performance, and propose enhancements to ensure scalability, redundancy, and security. They will also evaluate the network's effectiveness based on predefined business needs.

Student Responsibility

Students must:

1. Define their company’s business type, goals, and operational needs.
2. Justify their choice of network technologies based on business requirements.

Scenario: Choose Your Company

Imagine you have been hired as the lead network engineer for a company. You get to choose the type of company you are working for; Examples include but are not limited to:

- ✓ **E-Commerce Platform:** Handles online transactions and customer data.
- ✓ **Tech Startup:** Focused on product development with high bandwidth and collaborative requirements.
- ✓ **Educational Institution:** Serves staff, students, and visitors with diverse connectivity needs.
- ✓ **Logistics Company:** Requires seamless communication across warehouses and branches.
- ✓ **Healthcare Facility:** Needs high availability and secure patient data communication.

Your chosen company has three main locations:

Finance Subsidiaries Group – responsible for financial services that will support the company’s core business, this includes insurance, and investment management. (Accounting Unit, Treasury Unit, Budget Unit)

General Management – responsible for overseeing the overall operations, strategies, and the direction of the organization. (General Manager Office, Corporate Planning, Internal Auditing)

Corporate Support Group – ensures smooth operation of the corporation and enable the core functions to focus on company’s main objectives. (Human Resource, Administrative Unit, Legal Department)

Project Tasks

Part 1: Company and Network Requirements

1. Define Your Company (Be realistic)
 - Choose the type of company and provide a short description of its operations.
 - **National Development Company is a government-owned corporation under the Department of Trade and Industry (DTI), its main function is to promote industrial and economic development within the country. It has three main branch with different offices and departments that provides network resources for managing general management, corporate group and financial data.**
 - Describe unique networking requirements based on the company type (e.g., secure patient data for healthcare or scalability for a tech startup).

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GENDER SENSITIVENESS: During the examination and other aspects of instruction, the following will be observed:

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- The network is safe for company-wide communication, has improved scalability to support additional devices, and offers dependable access to shared resources including public computers, servers, and printers. The networking requirements include data security to help protect the important data especially the budget and financial branch inside the company.
- Set specific goals for network performance, scalability, and security.
 1. **Performance-Upgrade to higher-speed network connections**
 2. **Scalability- Transition to a fully virtualized network environment**
 3. **Security- Implement stronger authentication protocols**
- 2. Analyze Business Needs
Define:
 - IP addressing needs, including static IPs for servers and DHCP for client devices.
 1. **The business needs a more consistent addressing needs, hence the static IP addressing is implemented for more stability and easier configuration for the end devices.**
 2. **The devices that need the consistent IP addresses are given to all static IPs.**
 - Security concerns and solutions.
 1. **The lack of pass keys for the network, and the solution is the implementation of WPA2-PSK and passkey for each of the wireless access point within the network.**

Part 2: Network Design and Planning

1. Design the Network

Create a detailed topology showing:

- Three locations (Ex: HQ, Branch 1, Branch 2) connected over a WAN.

TREE TOPOLOGY

Finance Subsidiaries Groups (Accounting Unit, Treasury Unit, Budget Unit)

General Management (General Manager Office, Corporate Planning, Internal Auditing)

Corporate Support Group (Human Resource, Administrative Unit, Legal Department)

- Subnets using VLSM for efficient IP utilization.

1. Finance Subsidiaries Group – 50 host

Network ID : 192.168.1.0

Broadcast ID: 192.168.1.63

Host Range: .1 - .62

Subnet Mask: 255.255.255.192

2. General Management – 25 host

Network ID : 192.168.2.0

Broadcast ID: 192.168.2.31

Host Range: .1 - .30

Subnet Mask: 255.255.255.224

3. Corporate Support Group – 30 host

Network ID : 192.168.3.0

Broadcast ID: 192.168.3.31

Host Range: .1 - .30

Subnet Mask: 255.255.255.224

4. WAN #1 - Network ID : 192.168.4.0

Broadcast ID: 192.168.4.3

Subnet Mask: 255.255.255.252

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5. WAN #2 - Network ID : 192.168.4.4

Broadcast ID: 192.168.4.7

Subnet Mask: 255.255.255.252

- DHCP server placement with relay agents for branches.

2. Explain Your Design

Provide a written explanation of your design decisions, including:

- Why you chose your DHCP configuration.
- Why Static is suitable for your network.
 1. **For more reliable access, end devices like computers, laptop, cellular phones, and printers should be assigned IP addresses using the most popular static technique.**
- Security features implemented in your design.

Enabling WPA2-PSK and Pass Phrase for more secured wireless access points.

Part 3: Configuration and Implementation

1. Router and Switch Configuration

Configure basic settings (hostname, passwords).

Configuration of wireless access points like integration of passwords.

1 Security Configuration

Securing all the network devices using strong passwords and encrypted configurations.

Part 4: Advanced Testing and Analysis

1. Simulate and Test

Test the network under normal and failure scenarios:

- Simulate device or link failures to verify redundancy mechanisms.

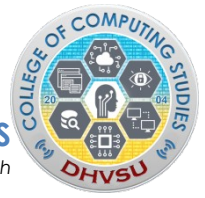
Ping Test: we ping the IP address of the other computer and packet loss to different endpoints within the network.
- Validate DHCP functionality at all locations.
- Test EIGRP convergence and failover effectiveness.

2. Document and Evaluate

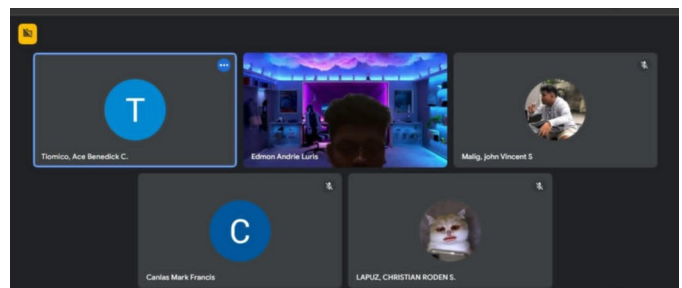
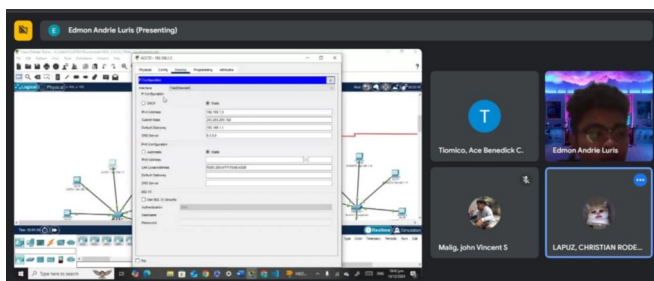
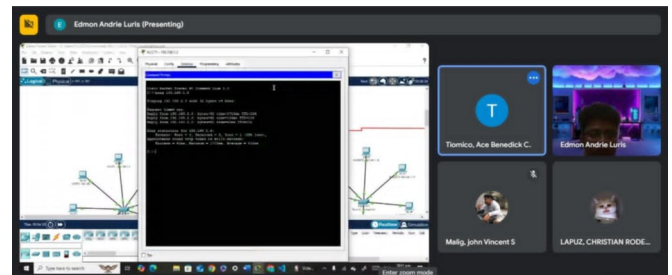
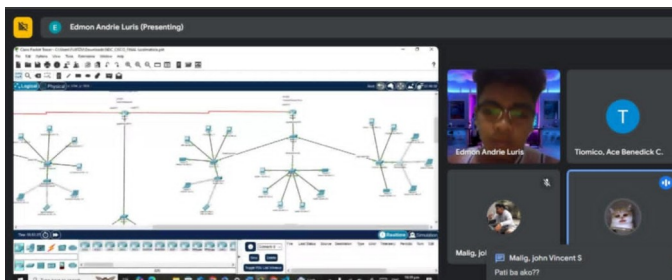
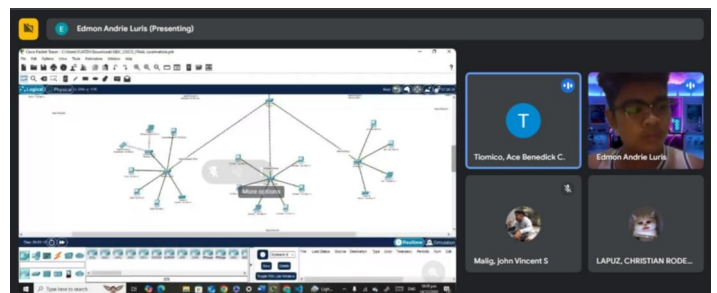
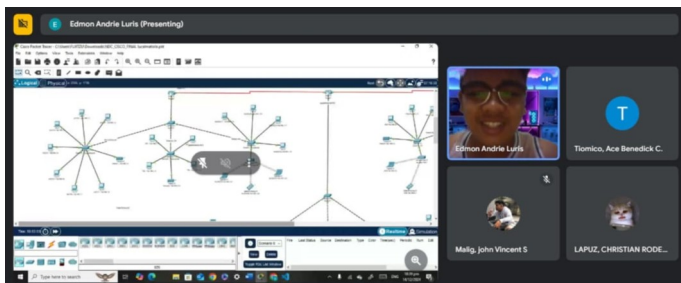
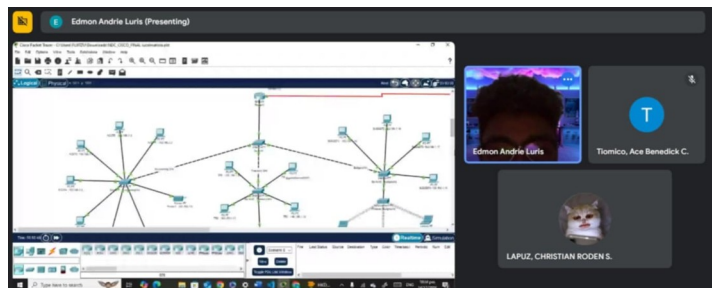
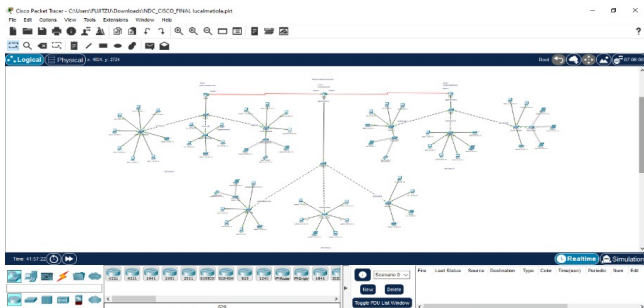
Provide a testing report with screenshots or output showing:

- Successful DHCP IP allocation.
- Connectivity between all devices.
- Documentation of the Group during the process of Project

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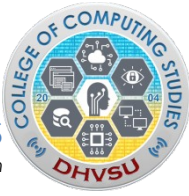


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Deliverables

- 1. **Company Description:** Explain your chosen business and its networking requirements.
- 2. **Network Design Document:** Include topology diagrams and an explanation of your design decisions.
- 3. **Configuration Files:** Provide full configurations for routers and switches.
- 4. **Testing Report:** Include results of connectivity tests, failover simulations, and performance analyses.

Project Grading Criteria

CRITERIA	Weightage
Company and Network Needs	15%
Network Design and Planning	25%
Configuration Accuracy	20%
Testing and Performance Analysis	25%
Documentation	15%